

Extended Abstract

Adapting to AI: How Information on Occupational AI Exposure Affects Educational Decisions*

Amanda Chuan

Germán Reyes

Joaquín Serrano

Hanzhe Zhang

May 2026

Abstract

We conduct a randomized information experiment with 3,841 undergraduates to study how AI affects college students' educational decisions. Treated students receive tailored, occupation-specific information about AI's labor market implications through an interactive visualization and a chatbot, while control students view format-identical placebo content. We track immediate belief updating with the survey, educational responses with student records, and post-graduation trajectories. In the survey, treated students lower their expected employment rate in their top-choice occupation by 2.4 to 3.3 percentage points at the 10- and 20-year horizons, raise the stated probability of changing major by 1.6 percentage points, and become 4.4 percentage points more likely to switch their top-choice occupation. Treatment also reframes major-choice criteria: treated students are 8 percentage points more likely to cite avoiding technological displacement as a reason for their major choice, a 35 percent increase relative to the control mean.

*Chuan: Michigan State University, School of Human Resources & Labor Relations (chuanama@msu.edu). Reyes: Middlebury College and IZA (greyes@middlebury.edu). Serrano: Universitat Autònoma de Barcelona (joaquin.serrano@uab.cat). Zhang: Michigan State University (hanzhe@msu.edu). We thank Christopher Boyette, Wenjia Cao, and Jie Wu for excellent research assistance. Approved by the MSU Institutional Review Board (STUDY202500204). Financial support from the Anthropic Economic Futures Program and Schmidt Sciences (AI at Work) is gratefully acknowledged.

1 Introduction

Generative artificial intelligence (AI) may reshape labor markets through channels distinct from prior automation waves. While previous technologies substituted for manual and routine tasks, contributing to a rising college premium (Autor et al., 2003; Goos et al., 2014; Goldin and Katz, 2008), generative AI excels at nonroutine cognitive tasks (Autor and Thompson, 2025; Eloundou et al., 2024). Early usage data suggest that adoption concentrates in cognitive-intensive fields: 37.2 percent of queries to one major AI system relate to computer and mathematical occupations, while only 0.1 percent fall in farming, fishing, and forestry (Handa et al., 2025). Several studies predict that generative AI will devalue some cognitive occupations while augmenting others (Brynjolfsson et al., 2025; Felten et al., 2023; Kogan et al., 2023; Pizzinelli et al., 2023), affecting college graduates differentially by their field of study. If these predictions hold, a student’s choice of major may matter more for labor market outcomes than in previous decades.

Whether students adjust their human capital investments in response to these shifts will shape how AI affects labor markets (Acemoglu and Autor, 2011; Autor, 2019). Traditional policy responses emphasize retraining displaced workers (Oreopoulos and Petronijevic, 2013), but retraining programs reach workers only after displacement, missing the chance to redirect investments earlier. College undergraduates are uniquely positioned to adjust their field of study before accumulating occupation-specific human capital that is costly to redirect (Altonji et al., 2012; Stinebrickner and Stinebrickner, 2014). Yet they may lack the information to do so. Many do not know which occupations face the greatest AI exposure, or how that exposure maps onto their choice of major.

In this paper, we conduct a randomized information experiment with 3,841 undergraduates at Michigan State University (MSU) to study how anticipated AI disruption affects college students’ educational decisions. Estimating these effects directly is challenging: AI adoption is widespread and endogenous, and students who use AI extensively differ from non-users in unobserved ways. We address this by varying students’ *information* about AI rather than AI use itself, building on a literature that uses information provision to generate experimental variation in contexts where randomizing the underlying economic variable is infeasible.¹

Students assigned to the treatment condition receive tailored, occupation-specific infor-

¹See Haaland et al. (2023) for a survey. Examples include information experiments that have identified causal effects of (perceived) returns to education (Jensen, 2010), tax incentives (Chetty and Saez, 2013), and college major choices (Wiswall and Zafar, 2015).

mation about AI’s labor market implications through two components: an interactive visualization that ranks occupations by AI exposure (measured using the Anthropic Economic Index, [Handa et al., 2025](#)), and a chatbot configured as a career advisor specializing in AI’s labor market impact. Students assigned to the control condition view a format-identical placebo: the same occupations sorted alphabetically, paired with a generic career-advisor chatbot prohibited from discussing AI. The two conditions are matched on format, interaction effort, and time-on-task; they differ only in AI-related content. We track immediate belief updating from the survey, actual educational responses from MSU student records collected 6 to 12 months post-intervention, and post-graduation labor market trajectories from individual-level Revelio Labs data on occupations, earnings, and education history.

We document four main findings from the post-treatment survey. First, treatment lowers students’ expected employment rate in their top-choice occupation at medium and long horizons, and to a smaller extent the subjective probability of personally being employed in that occupation. Second, treatment increases the stated probability of changing major and the probability that students switch their stated top-choice occupation within the survey. Third, treated students become more likely to cite avoiding technological displacement as a reason for their major choice and less likely to cite either job prospects or developing diverse skills. Fourth, despite reporting higher switching intentions, treated students do not engage more with the visualization and become less likely to express interest in exploring alternative majors with an advisor—suggesting that some respond to the information by retreating into greater commitment to their current path. Taken together, AI exposure information shifts both quantitative beliefs and the criteria students use to evaluate their educational decisions.

Our paper contributes to three strands of literature. First, we extend research on skill-biased technological change ([Autor et al., 2003](#); [Acemoglu and Autor, 2011](#); [Acemoglu and Restrepo, 2019](#)) by studying how *anticipated* AI disruption affects *ex-ante* human capital investments rather than *ex-post* labor market adjustments. Second, we contribute to the literature on information frictions in educational decision-making ([Wiswall and Zafar, 2015](#); [Zafar, 2013](#); [Fricke et al., 2018](#)) by examining whether information about a novel source of uncertainty—AI occupational exposure—can shift students’ beliefs, intentions, and choices. Third, we add to emerging work on the economic impacts of generative AI ([Brynjolfsson et al., 2025](#); [Noy and Zhang, 2023](#)) by documenting how individuals respond to AI-related labor market information and by isolating a framing channel that operates beyond standard belief updating.

2 Experimental Design

We recruited undergraduate students at MSU—a large public research university with approximately 49,000 students—through campus-wide email announcements that did not mention AI, to avoid selection on attitudes toward AI. Participants received a \$20 fixed payment for survey completion. The final sample comprises 3,841 respondents: 1,919 treatment and 1,922 control. Appendix Figure A4 provides an overview of the experimental design.

Treated students received two components conveying occupation-specific information about AI’s labor market effects. They first viewed an interactive visualization ranking approximately 490 occupations (searchable across over 5,000 O*NET titles) by AI exposure tier. The tiers are derived from the Anthropic Economic Index (Handa et al., 2025) and Brynjolfsson et al.’s (2025) estimates of how AI employment effects vary across exposure levels. Tiers were color-coded and displayed as bars of varying heights. Students selected occupations of interest, viewed tier-specific employment trend charts since November 2022, and explored occupations using filters for AI tier, occupation group, and education level (Appendix Figure A1). They then interacted with a Claude-powered chatbot configured as a career advisor specializing in AI’s labor market impact, with real-time access to occupation-level exposure data (Appendix Figure A2).

Control students received a format-identical placebo: the same occupations sorted alphabetically into four letter-range groups and displayed with uniform bars (Appendix Figure A3). The control chatbot acted as a generic career advisor and could not discuss AI. The two conditions are matched on format, interaction effort, and time-on-task; they differ only in AI-related content.

The survey collected pre- and post-treatment data on five sets of outcomes: expected employment rates and own employment probabilities for the top-choice occupation at multiple horizons; subjective earnings distributions; stated probabilities of changing major and applying to graduate school; major and occupation choices within the survey; and the factors influencing major choice. A voluntary, uncompensated offer to schedule a meeting with a career advisor served as a behavioral commitment outcome. We will link survey responses to MSU registrar records on major changes and course enrollments collected 6 to 12 months post-intervention, and to individual-level employment records from Revelio Labs.

We estimate intention-to-treat effects using:

$$Y_i = \alpha + \beta \text{InfoAI}_i + \mathbf{X}_i' \pi + \varepsilon_i,$$

where Y_i is the post-treatment outcome, InfoAI_i indicates treatment assignment, and $\mathbf{X}_i$ contains a progressively larger set of demographic and baseline-belief controls. We report estimates from no-controls and full-controls specifications. Standard errors are heteroskedasticity-robust throughout.

3 Preliminary Results

Table 1 confirms that randomization achieved balance across demographic characteristics and baseline beliefs; joint balance regressions yield p -values above 0.90. Table 2 reports the main treatment effects, organized into four families of outcomes: employment expectations, educational intentions and choices, reasons for choosing major, and engagement.

AI exposure information lowers students' expected employment in their top-choice occupation at medium and long horizons (Panel A of Table 2). The expected employment rate falls by 3.2 percentage points at the 10-year horizon and 3.1 percentage points at the 20-year horizon without controls (both $p < 0.01$). The estimates change little once we add baseline beliefs, demographics, and AI usage frequency: 2.4 percentage points at 10 years and 2.6 percentage points at 20 years (both $p < 0.01$). Effects on the subjective probability of personally being employed in the top-choice occupation are smaller but in the same direction: 1.3 percentage points at 10 years ($p < 0.05$) and 2.1 percentage points at 20 years ($p < 0.01$) with full controls. We find no consistent effects on salary expectations, perhaps because salary beliefs are more diffuse or less directly connected to occupation-level exposure information. The pattern suggests that treatment shifts beliefs about occupation-level employment more sharply than beliefs about students' own positioning within those occupations.

Treatment shifts both stated educational intentions and within-survey choices (Panel B). The stated probability of changing major rises by 1.2 percentage points without controls and 1.6 percentage points with controls ($p < 0.01$)—consistent with random variation being absorbed by pre-treatment beliefs. The stated probability of leaving college before graduating rises by 0.9 percentage points with controls ($p < 0.10$), suggesting that for some students the treatment raises uncertainty about whether their current educational path is worthwhile. Treated students are also 4.4 percentage points more likely to switch

their stated top-choice occupation within the survey ($p < 0.01$). Movement in stated occupation choices indicates that the treatment shifts not only point estimates but the choices students rank highest.

AI exposure information reframes the criteria students apply to their major choice (Panel C). Treated students are 8 percentage points more likely to cite avoiding technological displacement ($p < 0.01$)—a 35 percent increase relative to a control-group baseline of 22.6 percent—and 3 percentage points less likely to cite either job prospects or developing diverse skills ($p < 0.10$ for each). Beyond updating quantitative beliefs, the treatment makes technological displacement a salient consideration in major choice.

Engagement with the treatment materials is selective rather than uniformly higher (Panel D). Treated students do not spend more total time on the visualization, and they spend significantly less time on the selection page (-17.9 seconds, $p < 0.01$)—likely because the AI-tier sort makes the page easier to scan than alphabetical groupings. They spend slightly more time on the page that displays AI exposure tiers ($+3.5$ seconds, $p < 0.01$), consistent with selective attention to the AI-specific content. Despite reporting higher switching intentions in Panel B, treated students are 4.5 percentage points less likely to express interest in exploring alternative majors with an advisor ($p < 0.01$), and we find no difference in advisor-appointment requests. Some students may respond to the information by retreating into greater commitment to their current path—a pattern we will investigate in the planned heterogeneity analysis.

4 Next Steps

Three extensions are underway. First, we are implementing the pre-specified heterogeneity analysis, examining whether treatment effects differ by the AI exposure of the student’s baseline top-choice occupation, academic year, gender, declared-major status, and elicited risk and time preferences. We are most interested in heterogeneity by baseline exposure: students whose top-choice occupation has high AI exposure receive “bad news” from the treatment, while students with low-exposure occupations receive reassuring information. Second, we will link MSU administrative records to survey data in fall 2026, allowing us to measure actual major changes, course enrollment adjustments, and academic performance 6 to 12 months post-intervention. Third, linking survey responses to Revelio Labs employment data will let us trace the full chain from information provision to belief updating, educational adjustment, and labor market outcomes.

References

- Acemoglu, D. and D. Autor (2011). Skills, tasks and technologies: Implications for employment and earnings. In *Handbook of Labor Economics*, Volume 4, pp. 1043–1171. Elsevier.
- Acemoglu, D. and P. Restrepo (2019). Automation and new tasks: How technology displaces and reinstates labor. *Journal of Economic Perspectives* 33(2), 3–30.
- Altonji, J. G., E. Blom, and C. Meghir (2012). Heterogeneity in human capital investments: High school curriculum, college major, and careers. *Annual Review of Economics* 4(1), 185–223.
- Autor, D. and N. Thompson (2025). Expertise. Working Paper 33941, National Bureau of Economic Research.
- Autor, D. H. (2019). Work of the past, work of the future. In *AEA Papers and Proceedings*, Volume 109, pp. 1–32.
- Autor, D. H., F. Levy, and R. J. Murnane (2003). The skill content of recent technological change: An empirical exploration. *The Quarterly Journal of Economics* 118(4), 1279–1333.
- Brynjolfsson, E., B. Chandar, and R. Chen (2025). Canaries in the coal mine? six facts about the recent employment effects of artificial intelligence. Stanford Digital Economy Lab.
- Chetty, R. and E. Saez (2013). Teaching the tax code: Earnings responses to an experiment with EITC recipients. *American Economic Journal: Applied Economics* 5(1), 1–31.
- Eloundou, T., S. Manning, P. Mishkin, and D. Rock (2024). GPTs are GPTs: Labor market impact potential of LLMs. *Science* 384(6702), 1306–1308.
- Felten, E., M. Raj, and R. Seamans (2023). How will language modelers like ChatGPT affect occupations and industries? arXiv preprint arXiv:2303.01157.
- Fricke, H., J. Grogger, and A. Steinmayr (2018). Exposure to academic fields and college major choice. *Economics of Education Review* 64, 199–213.
- Goldin, C. and L. F. Katz (2008). *The Race between Education and Technology*. Cambridge, MA: Harvard University Press.
- Goos, M., A. Manning, and A. Salomons (2014). Explaining job polarization: Routine-biased technological change and offshoring. *American Economic Review* 104(8), 2509–2526.

- Haaland, I., C. Roth, and J. Wohlfart (2023). Designing information provision experiments. *Journal of Economic Literature* 61(1), 3–40.
- Handa, K., A. Tamkin, M. McCain, S. Huang, E. Durmus, S. Heck, J. Mueller, J. Hong, S. Ritchie, T. Belonax, K. K. Troy, D. Amodei, J. Kaplan, J. Clark, and D. Ganguli (2025). Which economic tasks are performed with AI? Evidence from millions of Claude conversations. Working paper, Anthropic.
- Jensen, R. (2010). The (perceived) returns to education and the demand for schooling. *The Quarterly Journal of Economics* 125(2), 515–548.
- Kogan, L., D. Papanikolaou, L. D. Schmidt, and B. Seegmiller (2023). Technology and labor displacement: Evidence from linking patents with worker-level data. Technical report, National Bureau of Economic Research.
- Noy, S. and W. Zhang (2023). Experimental evidence on the productivity effects of generative artificial intelligence. *Science* 381(6654), 187–192.
- Oreopoulos, P. and U. Petronijevic (2013). Making college worth it: A review of the research on the returns to higher education. *The Future of Children* 23(1), 41–65.
- Pizzinelli, C., M. Panton, M. Tavares, M. Cazzaniga, and L. Li (2023). Labor market exposure to AI: Cross-country differences and distributional implications. IMF Working Paper 23/216, International Monetary Fund.
- Stinebrickner, R. and T. R. Stinebrickner (2014). A major in science? initial beliefs and final outcomes for college major and dropout. *Review of Economic Studies* 81(1), 426–472.
- Wiswall, M. and B. Zafar (2015). Determinants of college major choice: Identification using an information experiment. *The Review of Economic Studies* 82(2), 791–824.
- Zafar, B. (2013). College major choice and the gender gap. *Journal of Human Resources* 48(3), 545–595.

Figures and Tables

Table 1: Baseline Balance

	Control	Treatment	Difference	<i>p</i> -value	Obs
<i>Panel A: Demographics</i>					
Female	0.474	0.487	−0.013	0.411	3,841
Age	20.25	20.20	0.047	0.317	3,841
White	0.571	0.595	−0.024	0.143	3,841
Black	0.058	0.060	−0.002	0.775	3,841
Hispanic	0.051	0.055	−0.004	0.557	3,841
Asian	0.123	0.119	0.004	0.743	3,841
Freshman	0.209	0.214	−0.005	0.703	3,829
Sophomore	0.239	0.245	−0.006	0.658	3,829
Junior	0.248	0.245	0.003	0.786	3,829
Senior	0.282	0.280	0.002	0.881	3,829
First-generation	0.204	0.200	0.004	0.807	2,933
Receives financial aid	0.753	0.739	0.014	0.401	2,936
Attended public HS	0.890	0.891	−0.001	0.919	2,894
<i>Panel B: Educational beliefs</i>					
Pr(change major) (%)	12.49	12.58	−0.098	0.885	3,784
Pr(attend graduate school) (%)	61.90	60.38	1.514	0.192	3,783
Pr(leave college) (%)	4.871	4.573	0.298	0.495	3,784
<i>Panel C: Employment beliefs</i>					
Emp. rate in top occ., current (%)	61.87	61.85	0.016	0.985	3,682
Emp. rate in top occ., 10 years (%)	66.51	65.66	0.851	0.298	3,679
Pr(employed, finish education) (%)	71.49	71.38	0.113	0.902	3,593
Pr(employed, 10 years) (%)	82.01	82.01	0.005	0.994	3,593
<i>Panel D: Major choice factors</i>					
Cites job prospects (baseline)	0.536	0.525	0.011	0.502	3,790
Cites avoiding tech. displacement (baseline)	0.228	0.225	0.003	0.838	3,790

Notes: Each row reports means by experimental arm and the *p*-value from a two-sample *t*-test of the difference. The total sample is 3,841 respondents (1,922 control, 1,919 treatment). Smaller samples in some rows reflect questions that appear later in the survey.

Table 2: Treatment Effects on Beliefs, Choices, and Engagement

Outcome	(1) No controls		(2) Full controls	
	Coeff.	SE	Coeff.	SE
<i>Panel A: Employment expectations (percentage points)</i>				
Employment rate in top occ., 10 years	−3.23	(0.90)***	−2.44	(0.57)***
Employment rate in top occ., 20 years	−3.12	(0.96)***	−2.53	(0.63)***
Pr(employed in top occ.), 10 years	−1.44	(0.85)*	−1.23	(0.53)**
Pr(employed in top occ.), 20 years	−1.83	(0.87)**	−2.02	(0.57)***
<i>Panel B: Educational intentions and choices (percentage points)</i>				
Pr(change major), stated	1.18	(0.75)	1.56	(0.48)***
Pr(leave college), stated	0.38	(0.60)	0.93	(0.49)*
Switched top-choice occupation in survey	4.62	(1.50)***	4.27	(1.50)***
<i>Panel C: Reasons for choosing major (percentage points)</i>				
Cites avoiding tech. displacement	7.90	(1.70)***	8.00	(1.70)***
Cites job prospects	−3.00	(1.70)*	−2.80	(1.60)*
Cites developing diverse skills	−3.00	(1.70)*	−3.30	(1.70)*
<i>Panel D: Engagement (sec. unless noted)</i>				
Wants to explore alt. majors (pp)	−4.70	(1.40)***	−4.50	(1.40)***
Visualization page 1 time	−17.92	(2.71)***	−17.92	(2.71)***
Visualization page 2 time	3.59	(1.39)***	3.48	(1.40)**

Notes: Each row reports a separate OLS regression of the outcome on an indicator for treatment assignment, with $N = 3,841$ except for items asked of subsamples (range: 2,943–3,841). Column (1) reports estimates with no controls (equivalent to a t -test). Column (2) reports estimates with the full control set: sex, age, race, year in college, first-generation status, financial aid eligibility, high school type, declared-major status, baseline beliefs about employment prospects and graduate school attendance, and self-reported frequency of generative AI use. “Switched top-choice occupation in survey” is an indicator equal to one if the student’s post-treatment top-choice occupation differs from the baseline. Heteroskedasticity-robust standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Appendix

Figure A1: Treatment Visualization: AI Exposure Information

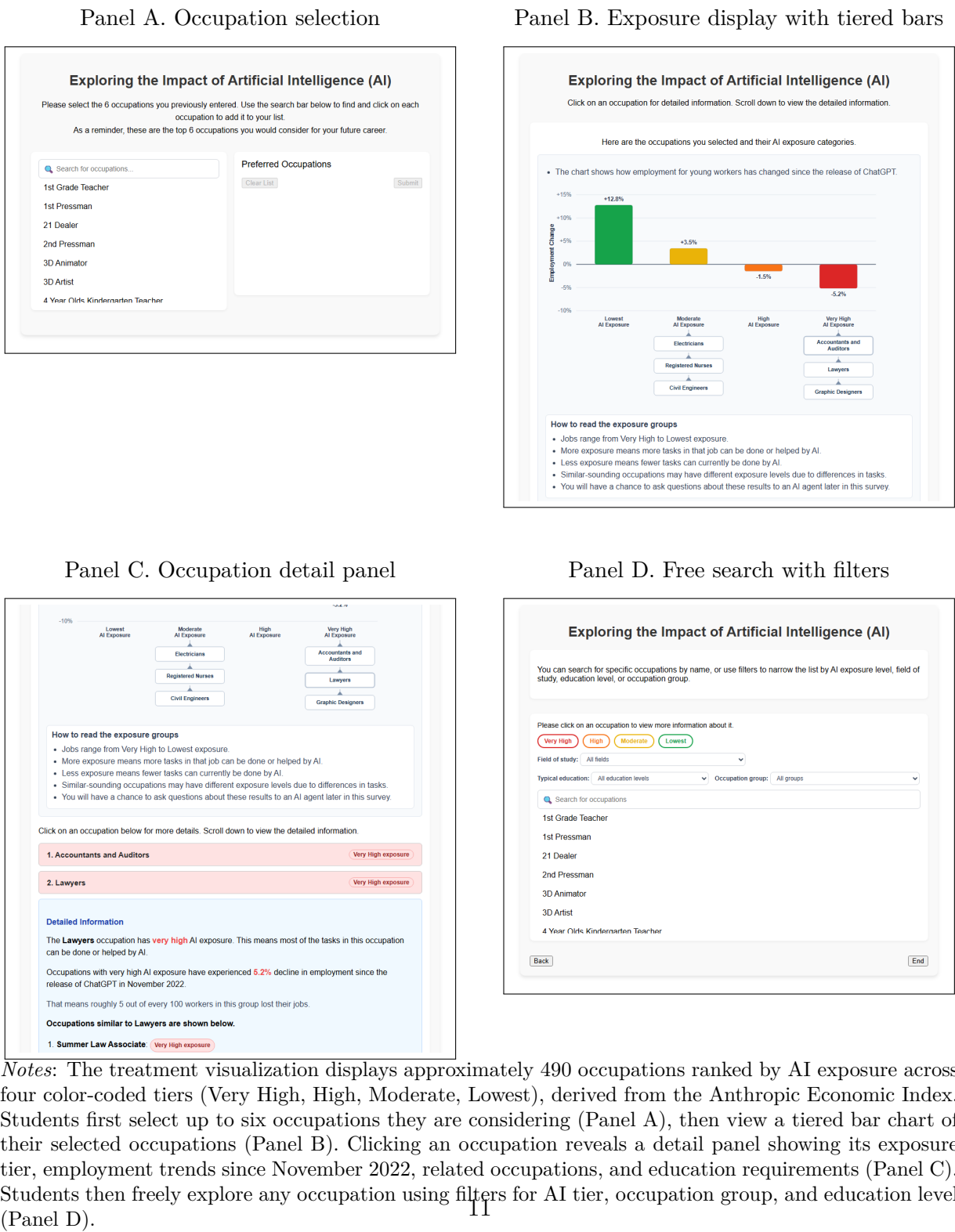
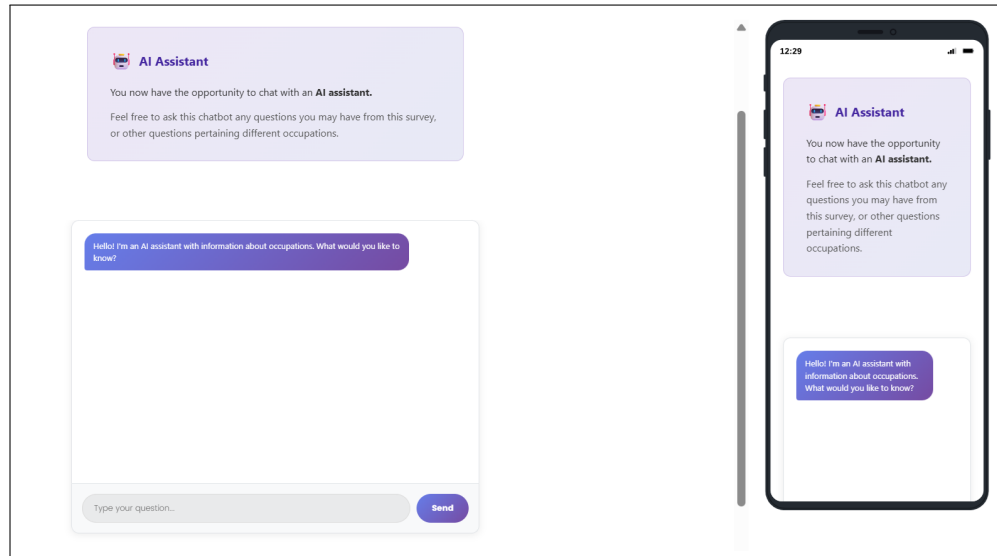


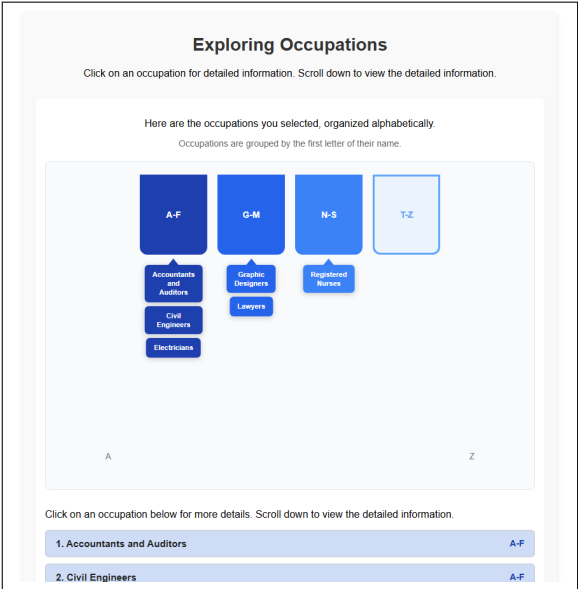
Figure A2: AI Chatbot Interface



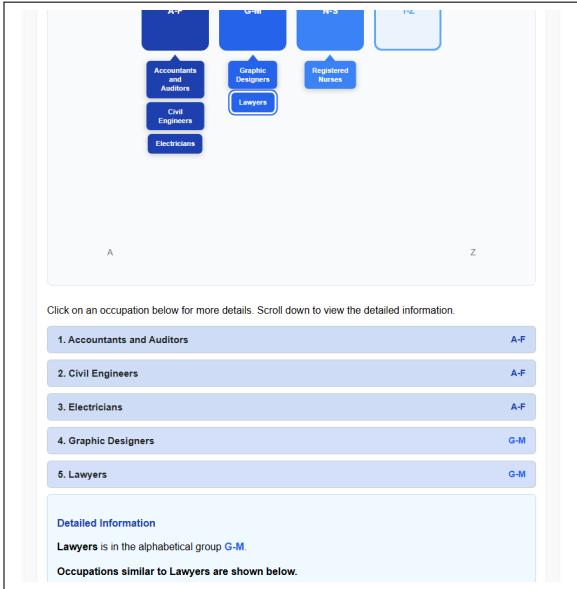
Notes: The AI chatbot interface as seen by treatment-group participants. The chatbot was powered by Claude (Anthropic) and acted as a career advisor specializing in AI's labor market impact, with real-time access to occupation-level AI exposure data. The control chatbot used the same interface but acted as a generic career advisor without access to any AI exposure information.

Figure A3: Control Visualization: Alphabetical Placebo

Panel A. Alphabetical display

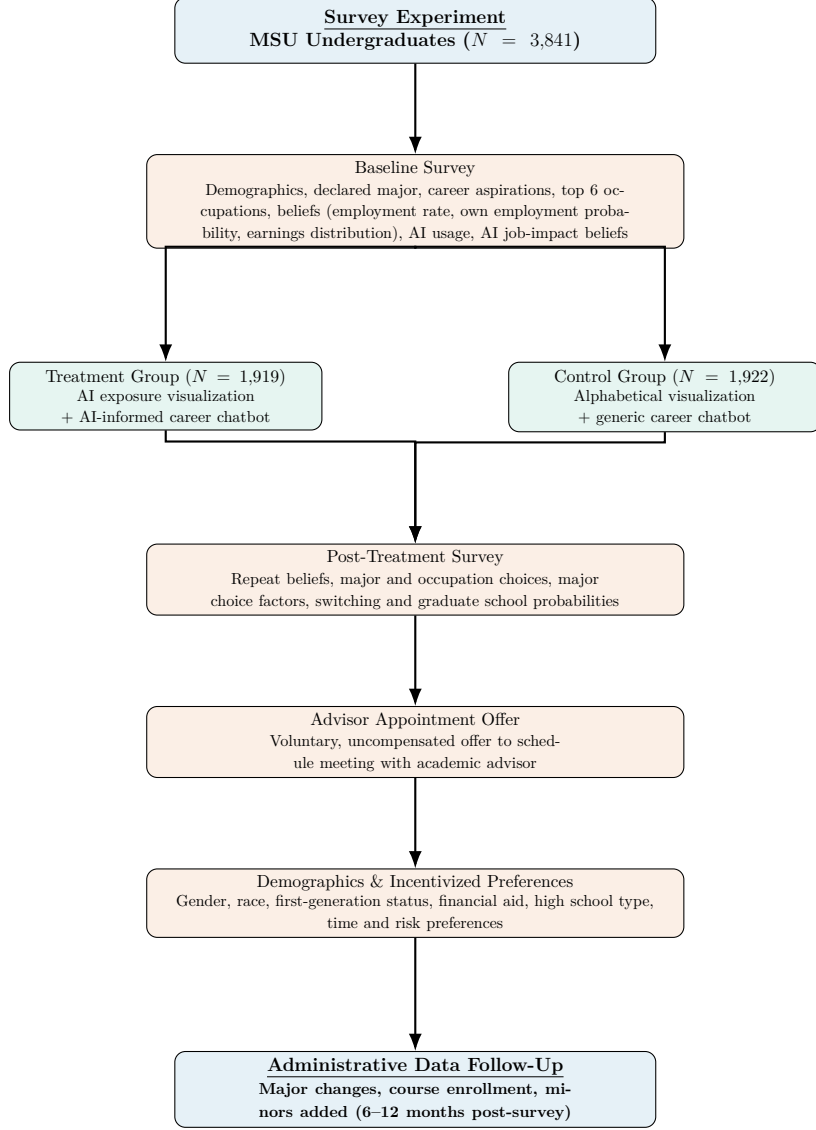


Panel B. Occupation detail panel



Notes: The control visualization displays the same occupations sorted alphabetically, grouped into four tiers by first letter (A–F, G–M, N–S, T–Z), with a uniform blue color scheme and equal bar heights. The detail panel shows related occupations and education requirements but no AI exposure data or employment trends.

Figure A4: Experimental Design Overview



Notes: Overview of the experimental design. All students complete the same baseline survey before randomization occurs. Treatment is assigned at the individual level with equal probability immediately before students view the visualization, so all pre-treatment responses are collected blind to assignment. Administrative data from the MSU Office of the Registrar will be collected 6 to 12 months after the survey.